6. Solution Requirements

Date	26 June 2026
Team ID	SWTID-2026-8135
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Not yet implemented as a login flow. A single default demo user ("user_default") is auto-seeded in users.json at startup by init_db().	Low
2	Authorization	All API and page routes are currently open; no role-based access control exists.	Low
3	External Interfaces	Integrates with the Google Gemini API (google-generativeai SDK) for chat, explanation, quiz, summary and recommendation generation, and with the HuggingFace MBZUAI/LaMini-Flan-T5 model (via transformers) for local concept explanation.	High
4	Transactions / Data Processing	Every request (chat, explain, quiz, summarize, recommend) is logged as a Query record and its AI output as a linked Response record, sharing a common QueryID (see main.py and utils/db_manager.py).	High
5	Reporting	No analytics dashboard yet; historical usage can be inspected directly from the JSON collections (queries.json, responses.json, quizzes.json, summaries.json, learning_paths.json).	Medium
6	Business Rules	If GEMINI_API_KEY is unset, invalid, or a live call fails, the system automatically falls back to a built-in demo/mock response generator (utils/gemini_service.py) so the UI never breaks.	High
7	Compliance to Laws / Regulations	No formal regulatory compliance (e.g., GDPR/HIPAA) is implemented — the project is scoped for academic / demo use.	Low
8	Other	Request payloads are validated with Pydantic schemas (models/schemas.py) — e.g., quiz count is constrained to 1–15 questions, explanation level to simple / medium / advanced.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Gemini calls use exponential backoff retries (up to 3 attempts) via call_gemini_with_retry(). Measured locally in demo/mock mode, average response times ranged 4–9 ms per endpoint (see Performance Testing section).	Demo mode: < 10 ms avg per request; live mode bounded by Gemini API latency + retries

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
2	Scalability	Single-process Uvicorn server with JSON flat-file storage (read-modify-write on every log call); not designed for high concurrency.	Suitable for demo / small classroom use; not yet load-tested for concurrent users
3	Security & Data Privacy	GEMINI_API_KEY is loaded from a local .env file (excluded via .gitignore); no encryption is applied to data stored in the JSON files.	API key never committed to source control
4	Reliability & Availability	Demo-mode fallback (get_mock_* functions) guarantees the UI keeps functioning even if Gemini is unreachable or unconfigured.	0 hard failures when GEMINI_API_KEY is absent, per current test suite
5	Usability & Accessibility	Responsive HTML/CSS frontend with a light/dark theme toggle (static/js/main.js) and dedicated pages per feature.	All 7 pages render successfully (verified by tests/test_api.py::test_read_pages)
6	Other — Maintainability	Each AI capability is isolated into its own router module (qna.py, explanation_module.py, quiz_module.py, summary_module.py, learning_path.py) with dedicated Pydantic schemas.	New AI capabilities can be added as a new router without touching existing modules